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Research paper

Dynamic security assessment of power systems using a deep learning and GAIN-based approach for addressing missing data

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ABSTRACT

The growth of artificial intelligence tools has sparked a surge in research attention to data-driven power system dynamic security assessment (DSA). Despite these advancements, DSA approaches face substantial challenges, including unintended data losses across communication, processing, and storage stages, as well as intentional data disruptions caused by cyberattacks. Such data issues can impair the effectiveness of DSA techniques and lead to erroneous decisions. This research introduces a novel hybrid data-driven approach for restoring missing data in phasor measurement units (PMUs) by the generative adversarial imputation network (GAIN) algorithm. It utilizes advancements in deep learning (DL) to develop a convolutional neural network (CNN) model for accurately assessing the dynamic security of power systems. The proposed model has been validated using the IEEE 39-bus and 118-bus networks, and the outcomes demonstrate the high accuracy and robustness of the proposed algorithm in achieving up to 99.48 % classification accuracy and maintaining stable performance even with up to 50 % missing data.

1. Introduction

The security of power systems is defined as their ability to sustain optimal performance under potential disturbances while ensuring uninterrupted satisfaction of consumer demands [1]. Dynamic Security Assessment (DSA) is a critical component of modern power system management, designed to evaluate a system's resilience against disturbances while ensuring stable operation during both transient and steady-state conditions. By analyzing the current operational state, DSA provides insights into the system's security status following anticipated events, empowering dispatchers to implement preventive measures to enhance reliability [2].

Traditional DSA methodologies encompass various approaches: (a) load flow calculations to identify voltage constraints or equipment rating violations, (b) time-domain simulations to analyze transient security and voltage stability, and (c) eigenvalue-based analysis for evaluating small-disturbance stability. These methodologies are further categorized into three key techniques: time-domain simulation (TDS), direct methods, and data-driven approaches. While TDS is accurate, it suffers from excessive computational demands, making it unsuitable for real-time applications. In contrast, direct methods leverage simplified models, leading to conservative or approximate results, but are similarly hindered by computational inefficiency and incompatibility with online applications [3]. The integration of renewable energy sources and demand response programs introduces significant uncertainty, further complicating the maintenance of system security and stability. These challenges underscore the need for advanced, real-time DSA methods capable of addressing modern power system complexities [4].

The advent of wide-area measurement systems (WAMS) and phasor measurement units (PMUs) has revolutionized DSA by enabling the application of artificial intelligence (AI) techniques. Data-driven models, leveraging machine learning (ML) and deep learning (DL), offer efficient solutions for real-time DSA by training on dynamic security datasets offline and applying these models to incoming measurements online [5]. Unlike traditional TDS or direct methods, these approaches significantly reduce computational burdens while improving accuracy and efficiency, making them well-suited for modern power system operations [6]. For example, Za'ter et al. introduced a Semi-Supervised Multi-Task Learning (SSMTL) framework, integrating conditional masked encoders and a topological similarity index. This method achieves a 200x speedup compared to traditional TDS methods, making it suitable for real-time applications [7]. Building on the need for adaptability in dynamic conditions, Ren et al. proposed a Reliable Transfer Learning (RTL) method that enhances pre-fault DSA by addressing unknown faults and incomplete data through adversarial

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Nomenclature

AI	Artificial Intelligence
BESS	Battery Energy Storage System
CNN	Convolutional Neural Network
DNN	Deep Neural Network
DSA	Dynamic Security Assessment
DER	Distributed Energy Resources
EMS	Energy Management System
ELM	Extreme Learning Machine
GAN	Generative Adversarial Network
GAIN	Generative Adversarial Imputation Network
GNN	Graph Neural Network
LSTM	Long Short-Term Memory
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
MSE	Mean Squared Error
PMU	Phasor Measurement Unit
ReLU	Rectified Linear Unit
RVFL	Random Vector Functional Link Network
SCADA	Supervisory Control and Data Acquisition
TDS	Time-Domain Simulation
TGAN	Temporal Generative Adversarial Network
VMD	Variational Mode Decomposition
WAMS	Wide-Area Measurement System

Mathematical Notations

X	Original data matrix
$\hat{X}$	Imputed data matrix
M	Binary mask matrix (indicating observed values)
Z	Noise vector for the GAN generator
$G(X,M,Z)$	Generator function in the GAIN model
$D(X,H)$	Discriminator function in the GAIN model
H	Hint matrix for guiding imputation
y_i	Actual class label
$\hat{y}_i$	Predicted probability for class iii
L	Loss function for CNN classification
$V(D,G)$	Objective function of GAIN
η	Learning rate for CNN training
W, b	Weights and biases in CNN layers
$\sigma(x)$	Activation function (e.g., ReLU, Softmax)
$MAPE$	Mean Absolute Percentage Error
$RMSE$	Root Mean Squared Error
$ADAI$	Average DSA Accuracy Index
Fc	Number of different contingencies
Am	Average accuracy of DSA with mmm missing PMUs

training and maximum classifier discrepancy, ensuring robust performance across diverse fault scenarios [8]. Complementing these advancements, an integrated scheme for online DSA employs Partial Mutual Information (PMI) and Pearson Correlation Coefficient (PCC) for effective feature selection, followed by Iterative Random Forest (IRF) to predict transient stability margins (TSMs) with high accuracy, demonstrating robustness to missing data, noise, and varying operating conditions in tests on IEEE 39-bus and 1648-bus systems [9].

To address data scarcity and topology changes, a DSA model combining TGAN-based data augmentation with CNN-based evaluation and adaptive batch normalization enhances performance on small datasets, improving generalization and accuracy for real-time DSA compared to traditional DL methods [10]. Similarly, an MLL-based DSA method leverages fault correlation to assess multiple power system faults simultaneously, achieving high accuracy in both sufficient and incomplete label scenarios, as validated on IEEE 39-bus and Illinois 200-bus systems [11]. In parallel, a data-driven scheme utilizing Sparse Projection Oblique Random Forests (SPORF) maps power system operating variables to security indices, enhanced by improved ADASYN for class imbalance and MRMR for feature selection, offering robust performance and adaptability to topology changes in tests on 23-bus and 16-bus systems [12]. Looking toward future grids with high renewable penetration, the FAST methodology employs machine learning and feature engineering, leveraging K-means clustering on normalized time series data to identify critical scenarios and detect voltage and frequency limit violations, as demonstrated on the 2028 WECC 22,000+ bus system [13]. Collectively, these advancements highlight the transformative potential of AI-driven DSA in ensuring reliable, real-time power system stability.

Voltage stability, a critical factor in preventing blackouts and equipment damage, has also benefited from data-driven DSA advancements. By incorporating voltage stability indices, these methods enable early detection of instability risks, allowing operators to take timely preventive actions. For example, Azad et al. proposed the S-Index, a novel real-time voltage stability indicator for multi-line and n-bus systems, integrating transmission and distribution systems to achieve higher accuracy in identifying voltage collapse points [14].Furthermore, advanced control and optimization techniques, such as sliding mode and fractional-order controllers, have been applied to hybrid interconnected systems to improve dynamic behavior under nonlinear conditions [15]. Similarly, metaheuristic optimization strategies have enhanced protection coordination in microgrids, supporting the shift toward intelligent, adaptive frameworks for system-wide security [16].These developments support the ongoing shift toward intelligent, adaptive frameworks for system-wide security assessment.

The increasing reliance on PMUs introduces challenges related to data incompleteness due to communication congestion, hardware failures, transmission delays, and cyberattacks. Addressing these vulnerabilities requires two complementary strategies: (1) developing robust DSA models capable of accurate forecasting despite incomplete data and (2) integrating effective data recovery mechanisms into the DSA framework.. Liu et al. proposed an integrated online DSA framework using PMI and PCC for key variable selection, followed by IRF for TSM prediction, enhanced by spatial-temporal visualization for real-time monitoring [9,17]. A notable contribution by Zhang et al. introduced a robust ensemble data analytics model for incomplete PMU measurements. This model clusters PMUs to maintain system observability and ensure accuracy, even under significant data loss [18]. Ren et al. further advanced this field with a super-resolution perception-based incremental learning approach, enhancing DSA robustness to data loss [19]. A noteworthy approach is the application of Generative Adversarial Networks (GANs) for missing data recovery, as demonstrated by Ren and Xu [20],This data-driven solution effectively fills missing PMU measurements without relying on system topology, thus enhancing the reliability of DSA models. Emerging techniques further enhance the resilience of DSA to missing data. Additionally, Cheng et al. developed a deep learning algorithm based on a Seq2Seq LSTM architecture, incorporating prior knowledge matrices and decoupling PMU data into magnitude and trend components [21]. The Generative Adversarial Imputation Network (GAIN) further excels in imputing missing data with high accuracy and robustness, offering a probabilistic approach suitable for real-time DSA [22]. Pei et al. introduced a Two-Stage Denoising Diffusion Model (TSDM) that integrates anomaly detection with classifier-guided diffusion and Bayesian optimization, outperforming traditional Kalman filters in accuracy and efficiency [23]. Addressing the lack of training data in DSA, Bogodorova et al. utilized Variational Mode Decomposition (VMD) for data augmentation, preserving oscillatory patterns and improving DNN performance in small

signal stability tasks [24]. Other studies have explored adaptive recovery methods using transfer learning to handle topology changes and unlearned faults [25], while a deep learning framework integrating Conditional Tabular GANs (CTGAN), Graph Convolutional Networks (GCN), and Maximum Mean Discrepancy (MMD) addresses database imbalance and topology changes for pre-fault DSA [26]. Shang et al. propose a hybrid model for imputing missing traffic flow data, using k-means clustering to group road segments by traffic patterns, capturing spatial correlations. LSTM networks with an attention mechanism extract temporal dependencies, while AdaBoost integrates these networks to enhance the accuracy of imputation. Validated on the PeMS dataset, the model achieves stable and accurate imputation for 10–60 % missing data across various scenarios [27,28] proposed a novel deep learning architecture for distribution system topology identification, effectively handling missing PMU measurements to enhance state estimation accuracy. Tested on IEEE 13 and 37-bus systems, their model demonstrates robustness to data loss, supporting reliable real-time DSA applications.

Various deep learning and machine learning models, such as SVM, Random Forest, RNN, and GNN, have been applied to DSA tasks. Among these, CNNs are particularly effective for time-series classification due to their ability to extract hierarchical features with lower computational cost and faster convergence. They also show strong resilience to noise and missing data, making them well-suited for PMU-based assessments. Prior studies [10,26] support the use of CNNs in high-dimensional power system data analysis.

Traditional imputation techniques such as mean substitution, k-NN, and MICE have been widely used for handling missing data, however, they face significant limitations when applied to complex and high-dimensional time-series data like PMU measurements. The k-NN method imputes missing values by averaging the nearest neighbors, which can lead to reduced accuracy in the presence of nonlinear relationships or temporal dependencies [29]. Similarly, MICE assumes that data is missing at random (MAR)—an assumption that may not hold in real-world power systems affected by cyberattacks or communication failures. MICE also struggles to model nonlinear dependencies and becomes computationally intensive as the dataset grows in size and complexity [30]. Even some deep learning-based methods, such as autoencoders and LSTMs, may suffer from instability or require extensive tuning, particularly under high missing data rates [31]. In contrast, GAIN adopts a probabilistic, adversarial learning approach that models the true distribution of the data and generates realistic imputations without needing prior knowledge of the system topology. Its ability to handle nonlinearities, adapt to high levels of missingness, and operate efficiently makes it especially well-suited for real-time and resilience-focused applications like dynamic security assessment in modern power systems.

Although data-driven methods have offered advantages in real-time DSA compared to traditional methods, A new DL-based method was needed to address some issues, including a robust DSA model like CNN, which performs more accurately than simple machine learning algorithms like support vector machine (SVM) classification methods, even in large-scale power systems. CNNs identify and extract essential internal structures from inputs through convolution and pooling operations, generating more informative and robust features. In addition, this robust DSA model is integrated with a missing data recovery model (GAIN), which has shown promising results even in cases with a high rate of incomplete data, performing remarkably well for real-time DSA.

This paper proposes a novel hybrid deep learning framework that integrates the Generative Adversarial Imputation Network (GAIN) with a CNN-based DSA model to address the critical challenge of missing data in power system security assessment. Unlike traditional imputation methods, the GAIN component reconstructs missing PMU data with high accuracy, independent of power grid topology, ensuring robust performance across diverse network configurations. The CNN-based DSA model leverages active power and rotor angle data from system generators to deliver precise security assessments, distinguishing secure from insecure network states even under significant data loss. This framework offers several key advantages: it achieves superior predictive accuracy compared to conventional ML methods like SVM, maintains robustness in high-missing-data scenarios, and scales effectively to large systems. Validated on the IEEE 39-bus and IEEE 118-bus systems, the proposed approach demonstrates exceptional reliability and adaptability, enabling system operators to implement timely preventive control measures. The salient contributions of this work are:

1. **Robust Handling of Missing Data:** The framework effectively manages simultaneous missing data from multiple PMUs over consecutive time intervals, maintaining high DSA accuracy even under substantial data loss.
2. **Topology-Independent Data Recovery:** The GAIN algorithm's ability to impute missing data without relying on system topology enhances its applicability across diverse power system configurations, offering a significant improvement over topology-dependent methods.
3. **High Predictive Accuracy with CNN**: The CNN-based DSA model extracts critical features from high-dimensional PMU data, outperforming simpler ML models like SVM by generating robust and informative features through convolution and pooling operations.
4. **Scalability and Versatility:** Extensive validation on the IEEE 39-bus and IEEE 118-bus systems demonstrates the framework's scalability, ensuring reliable performance across a wide range of system sizes and operating conditions.
5. **Comprehensive Sensitivity Analysis:** A rigorous sensitivity analysis of the GAIN algorithm under varying levels of missing data confirms its robustness, providing confidence in its real-time applicability for DSA.

The remainder of this paper is organized as follows: Section II details the proposed framework, including the CNN-based DSA model and the GAIN-based missing data recovery model. Section III presents the dataset creation process, parameter selection for the deep learning algorithms, and the performance evaluation of the proposed approach. Section IV is written to conclude the article.

2. Methodology

This section presents the proposed hybrid deep learning framework for Dynamic Security Assessment (DSA), integrating the Generative Adversarial Imputation Network (GAIN) for missing data recovery with a Convolutional Neural Network (CNN) for operational state classification. The framework leverages active power and rotor angle data from phasor measurement units (PMUs) to achieve robust and accurate DSA, even under significant data loss.

2.1. Generative adversarial imputation network

GAIN, as a novel advancement in GAN technology, demonstrates remarkable success in operating effectively even in situations where complete data is not accessible [15]. GAIN models have emerged as promising unsupervised deep learning algorithms for dealing with the issue of missing data in power systems. These models can undergo training to estimate missing values, detect outliers, and fill in missing data [14]. As illustrated in Fig. 1, a GAIN architecture comprises three interconnected deep learning models: a generator (G), a discriminator (D), and a hint generator (h).

The G is responsible for generating data samples that follow the distribution observed in the historical training data. In simple terms, the G is trained to learn the G(z) mapping function, seeking to map a random noise vector z to the actual data. Therefore, the G model aims for its produced data to be securely passed via the D model and obtain the actual data label (Defeat the D). The D model seeks to identify any data

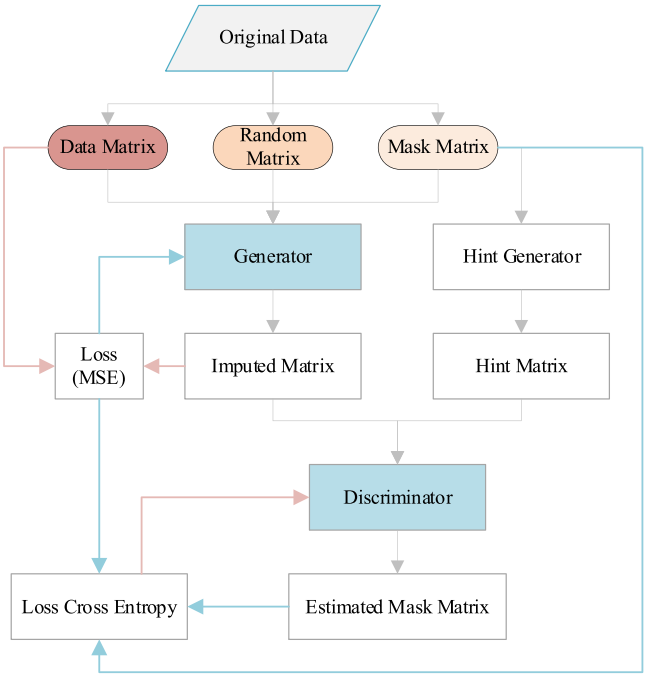


Fig. 1. The GAIN missing data algorithm framework.

originating from G as counterfeit, regardless of its quality. The hint matrix guarantees that the G produces samples that accurately reflect the underlying data distribution. The Hint Mechanism (H) provides partial information about which components of the data were originally observed, guiding the discriminator during training.

The iterative training process of models G and D is carried out until the generated data by model G closely resembles the historical data, making it increasingly challenging for model D to differentiate between real and generated data. The generator(G) takes input vectors. $\widehat{X}$, M, and a noise variable, Z, to produce an output$\overline{X}$, a vector of imputations. The function G is defined as $G:\widetilde{\chi}\times\{0,1\}^d\times[0,1]^d\rightarrow\chi$, where $Z=(Z_1,\ldots,Z_d)$represents d-dimensional independent noise. It is then defined that the random variables$\overline{X},\widehat{X}\in\chi$.

$$\overline{X}=G(\widetilde{X},M,(1-M)\odot Z) \tag{1}$$

$$\widehat{X}=M\odot\widetilde{X}+(1-M)\odot\overline{X} \tag{2}$$

$\odot$denotes element-wise multiplication. $\overline{X}$is the vector of imputed values (even for observed instances) and the completed data vector. This means replacing each * in the partial observation with the corresponding value$\overline{X}$.To explain hints, a hint mechanism refers to a random variable with values in a space$H$, both of which are defined. H depends on M; for each (imputed) sample$(\widehat{x},\ m)$, h will be drawn according to the distributionH|M $=m$. h is passed as an additional input to the discriminator, transforming it into a function$D:\chi\times H\rightarrow[0,1]^d$. In this context, the i th component represents the probability that the i th element $\widehat{X}$was observed, conditional on $\widehat{X}=\widehat{x}$andH $=h$.

D is trained to maximize the probability of accurately predicting M, and G is trained to minimize the likelihood of M being predicted by D. The quantity V(D, G) is defined as Eq (3), where "log" denotes element-wise logarithm, and the dependence on G is through$\widehat{X}$.

$$V(D,G)=\left[E_{\widehat{X},M,H}\left[M^T\log D(\widehat{X},H)+(1-M)^T\log(1-D(\widehat{X},H))\right]\right] \tag{3}$$

The objective of GAIN is defined as the minimax problem given by:

$$\min_G\max_D V(D,G) \tag{4}$$

The loss function $L:\{0,1\}^d\times[0,1]^d\rightarrow\mathbb{R}$ is defined as:

$$L(a,b)=\sum_{i=1}^{d}\left[a_i\log(b_i)+(1-a_i)\log(1-b_i)\right] \tag{5}$$

Expressing (5) can then be rewritten as:

$$\min_G\max_D \mathrm{E}[L(\mathrm{M},\widehat{\mathrm{M}})] \tag{6a}$$

Training is performed iteratively in mini-batches. During each iteration:

1. The generator imputes missing values using observed data and random noise.
2. A hint matrix is sampled and provided to the discriminator.
3. The discriminator attempts to classify whether each value in $\widehat{X}$ is real or imputed.
4. The generator and discriminator are updated in alternating steps using backpropagation.

2.2. CNN algorithm

Although data classification is integral to data mining, the substantial size and complexity of modern datasets continue to pose significant challenges. Deep learning methodologies, particularly convolutional neural networks (CNNs), have been investigated as alternatives to traditional feature-based approaches. By automatically identifying and extracting intrinsic structures from input data, CNNs present a promising framework for data classification [19]. A CNN architecture typically involves two core components: convolutional and pooling layers for deep feature extraction from raw data, followed by a multilayer perceptron (MLP) for classification. The layers comprising this architecture are outlined below [19]:

1. Input Layer: Comprising neurons, where k represents the number of input data series, and N represents the length of each univariate data sequence.
2. Convolutional Layer: Convolution filters are used to perform convolution operations on the data from the previous layer. The selection of parameters—such as the number of filters (m), convolution stride (s), and filter size (l)—should be guided by domain knowledge or experimental findings. In this case, k denotes the number of data variables in the previous layer, while l indicates the filter's length. Additionally, determining a nonlinear transformation function, f, is necessary. For example, if the previous layer provides k-variate data, with each univariate data sequence having length N, then after the convolution operation, m-variate data are obtained. The size of each univariate sequence would be $\lfloor(N-l)/s+1\rfloor$.
3. Pooling Layer: Pooling involves dividing a feature map into N equal-length segments. Each segment is represented by either its average or maximum value. This operation down-samples the convolutional outputs, reducing variability in hidden activations and, in turn, enhancing their consistency and reliability [20].
4. Feature Layer: The original data is transformed into a series of feature maps by applying multiple convolution and pooling operations. These feature maps are then combined to produce a new, comprehensive long data representation that reflects the original input within the feature layer.
5. Dropout Layer: Dropout is a regularization technique in deep learning that prevents overfitting. It randomly eliminates neurons with a probability of p or retains them with a probability of (1–p), thereby improving network performance and reducing the risk of overfitting.
6. Dense Layer: The fully connected layer is commonly employed in neural networks. In this layer, each neuron connects to all neurons in the previous layer. It is often used in the final stages of a network to

transform extracted features into a representation more conducive to classification.

7. Output Layer: The output layer of the neural network architecture comprises n neurons, where n represents the number of data classes. This layer is fully connected to the feature layer. In classification tasks, the most common method involves selecting the output neuron with the highest activation value and using it as the corresponding input data class label.

The CNN is trained using a sequence of training samples. The steps to train a network on a multivariate or univariate data target output vector (y_t) are described as follows:

Training begins with initialization. First, the CNN architecture is determined, typically comprising two convolutional layers and two pooling layers. The number of neurons in the input and output layers is configured based on the classification task. Next, all CNN parameters, including weights and biases, are initialized with small random values. Finally, a learning rate (η) and an activation function (f) are selected. The ReLU function is a commonly used activation function.

$$f(x) = \text{ReLU} = \max(0, x) \tag{6b}$$

Next, a training sample is randomly drawn from the training set, and the output for each layer is computed sequentially. The output of the convolutional layer is expressed as:

$$C_r(t) = f\left(\sum_{i=1}^{l}\sum_{j=1}^{k} x(i + s(t-1), j)\omega_r(i,j) + b(r)\right) \tag{6}$$

$x_t \in \mathbb{R}^{N\times k}$represents the input data or the output from the preceding layer, S denotes the convolution stride, signifies the ith component of the rth feature map, and $\omega_r \in \mathbb{R}^{l\times k}$ corresponds to the weights and bias of the rth convolution filter. The pooling layer's output is given as:

$$P_r(t) = g(C_r((t-1)l+1, C_r(t-1)l+2), \ldots, C_r(tl)) \tag{7}$$

The pooling strategy, represented by function g, is a common technique in data analysis. The two most prevalent pooling methods are averaging and max pooling. This process reduces data points while maintaining the same number of feature maps. The output layer's result can be expressed as:

$$O(j) = f\left(\sum_{i=1}^{M} z(i)\omega_f(i,j) + b_f(j)\right), j = 1, 2, \ldots, n \tag{8}$$

The Eq. (9) provided is used to calculate the mean-square error. It involves several variables, such as the final feature map denoted by z in the feature layer, the bias of the output layer represented byb_f, and the connection weights between the feature layer and the output layer are characterized by $\omega_f \in \mathbb{R}^{M\times n}$.

$$E = \frac{1}{2}\sum_{k=1}^{n} e(k)^2 = \frac{1}{2}\sum_{k=1}^{n} (O(k) - y(k))^2 \tag{9}$$

Then, the weights and biases will be updated by the gradient descent method. Where p is the parameter's value, and p refers to this CNN.

$$p = p - \eta\frac{\delta E}{\delta p} \tag{10}$$

According to the CNN algorithm's structure, the proposed model has three CNN layers, three 1D Max Pooling layers, four dropout layers with a 25 % rate, and two dense layers, as shown in Fig. 2.

2.3. Classification rules

Dynamic simulation data (RMS Simulation) using the rotor angle criterion, work points are divided into two secure and insecure states. If all the differences between the two generator rotor angles are <180 degrees during the simulation time after the fault occurs, that point is labeled as secure$Y_i = 1$; Otherwise, it is labeled as insecure$Y_i = 0$.

2.4. Framework for dynamic security assessment with GAIN-based data recovery

Dynamic security assessment (DSA) in modern power systems necessitates robust frameworks that effectively handle incomplete data scenarios, particularly in encountering communication failures, cyber threats, and measurement inaccuracies. The proposed framework integrates **a** GAIN model for missing data recovery with a deep learning-based CNN for dynamic security classification. This hybrid approach enhances real-time decision-making capabilities while ensuring robustness against data loss, making it suitable for practical deployment in operational environments. The framework consists of three key phases depicted in Fig. 3: an offline training phase and an online security assessment phase. These phases work in tandem to provide real-time security assessments while mitigating the impact of missing phasor measurement unit (PMU) data. For dataset generation, time-domain simulations are conducted on standard power system models such as the IEEE 39-bus and 118-bus test systems. These simulations cover various system operating points, including secure and insecure states, and incorporate different fault scenarios and contingencies. Data is then used to train the GAIN model for missing data recovery and the CNN model for dynamic security classification.

2.4.1. Offline training phase

The offline training phase is a preparatory step in which models are trained using a dataset to ensure high accuracy in both missing data imputation and DSA classification algorithms. At first, the comprehensive datasets are generated using RMS simulation under various contingencies. These datasets include secure and insecure OPs derived from rotor angle stability analysis. Then, the GAIN model, a specialized generative adversarial network (GAN) designed for missing data imputation, is trained using a dataset with varying degrees of missingness. The training process optimizes the generator and discriminator networks iteratively until the generator reliably recovers missing PMU data with minimum error. In other words, The generator produces data to fill in missing values, while the discriminator assesses the accuracy of the imputed values based on historical data patterns. Afterward, the CNN-based DSA model is trained using complete datasets, the architecture consists of Convolutional, Pooling, and Softmax output layers and classifies system states as secure or insecure based on learned patterns in the training process using the categorical cross-entropy loss function.

2.4.2. Online security assessment phase

Once trained, the framework transitions to online classification, where incoming PMU data is continuously processed to assess dynamic security. At first, Real-time PMU measurements are received to be analyzed for completeness. If no data loss is detected, the measurements are directly input into the CNN-based DSA model. The data is passed to the GAIN model for imputation if missing values are identified. The GAIN model recovers the incomplete dataset. The generator produces missing values based on learned patterns, while the discriminator

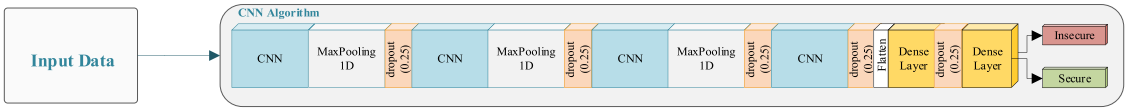


Fig. 2. Structure of the proposed CNN model.

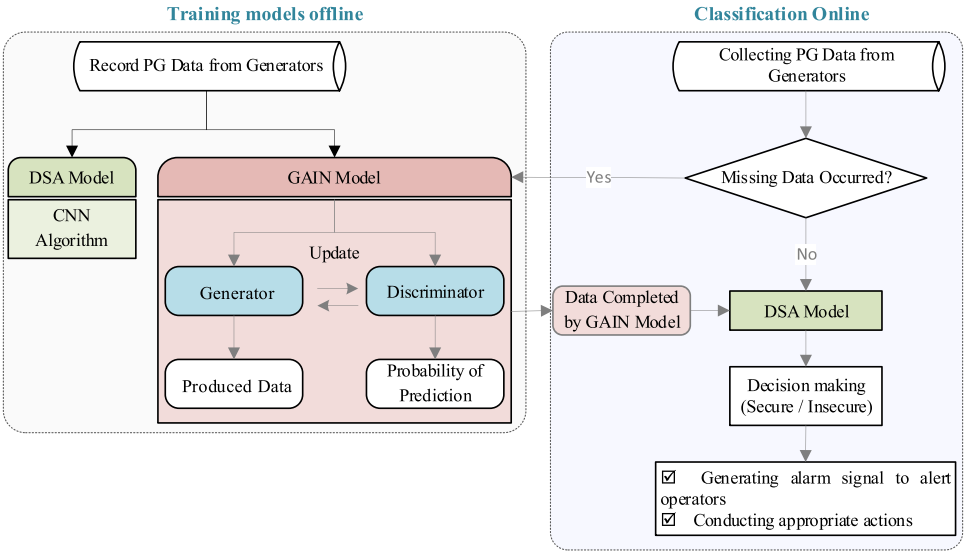


Fig. 3. Structure of the proposed security assessment model.

differentiates between real and imputed data. The corrected dataset is then passed to the CNN classifier for security assessment. Then, the CNN model classifies the system state as secure or insecure based on the processed dataset. If an insecure state is detected, preventive control measures are triggered to mitigate potential stability risks.

2.4.3. Performance evaluation criteria

2.4.3.1. MAPE. The main absolute percentage error (MAPE) is a commonly used metric for assessing prediction accuracy. The GAN's ability to fill missing data is tested at 5 % and 10 % rates, evaluated by mean average percentage error (MAPE) [32].

$$MAPE = \frac{1}{N_t}\cdot\sum_{i=1}^{N_t}\left|\frac{y_i^{real} - y_i^{predicted}}{y_i^{real}}\right| \tag{11}$$

N_t Denotes the total number of testing instances, y^{real} is the real data value, and $y^{predicted}$is the predicted missing data value.

2.4.3.2. RMSE. Root Mean Squared Error (RMSE) is often used to evaluate the imputation quality in continuous datasets. RMSE quantifies the square root of the average squared differences between the actual and predicted values [33]. The RMSE formula is given by:

$$RMSE = \sqrt{\frac{1}{N_t}\cdot\sum_{i=1}^{N_t}(y_i - \hat{y}_i)^2} \tag{12}$$

2.4.3.3. ADAI. This paper uses the average DSA accuracy index (ADAI) to quantify DSA accuracy and robustness in incomplete PMU measurements.

$$ADAI = \frac{1}{M}\frac{1}{F_c}\sum_{f=1}^{Fc}\sum_{m=1}^{M}A_m \times 100\% \tag{13}$$

F_c represents the number of different contingencies. A_m is the average accuracy of DSA with m missing PMUs, while ADAI is the average DSA accuracy when all PMUs are missing.

3. Numerical results

Training the Dynamic Security Assessment (DSA) model involves using Python version 3.9.7, Keras, and TensorFlow. Additionally, PyTorch trains the missing data recovery model (GAIN) in Python software. These simulations are carried out on a laptop with a 1 GHz five-core processor and 8 GB of RAM. The Digsilent generates the database, producing numerous scenarios for the 39-bus and 118-bus IEEE system.

3.1. Dataset

To evaluate the proposed dynamic security assessment (DSA) framework, we used two benchmark power system datasets: the IEEE 39-bus test system, as shown in Fig. 4, and the IEEE 118-bus test system. These datasets provide comprehensive data that reflect the operational characteristics of typical power systems. The dataset includes root mean square (RMS) simulation results for various operating points, each corresponding to different system loads and fault scenarios. These operating points are generated by simulating a range of three-phase faults and system contingencies. The study's Contingencies involve ten types of three-phase faults that occur and remain fixed for 0.12 s. These Contingencies are listed in Table 1 10,000 instances were produced, with recorded data on generator power and rotor angle. In the Dataset, 35 % of operational points are insecure, and 65 % are secure.

Creating the 118-bus IEEE system dataset followed a process similar to the 39-bus system. Table 2 presents the list of 3-phase faults. For further details, including schematics and system specifications, please refer to the provided reference [34].

Using Eq. (14) to construct the database, operating points are generated according to each bus's expected load demand level for different system loads.

$$\rho_{Li} = n\rho_{Li}^0 \pm \sigma \tag{14}$$

The load factor of the power system (ρ_{Li}^0) is a random number between 0.7 and 1.25; n is equal to 1 or −1, and the value is 0.08.

The RMS simulation generates data for a range of secure and insecure operating points. For each operating point, a three-phase fault is applied at one of the predefined buses (see Tables 1 and 2), lasting 0.12 s before line tripping. The system response is then simulated for 10 s using DIgSILENT PowerFactory. Load variability is introduced by modifying the base case loading profile using random scaling factors drawn from a uniform distribution in the range [0.7, 1.25], as described in Eq. (14).

To evaluate the proposed method under missing data conditions, random missing data simulations were introduced at varying levels (5 %, 10 %, 20 %, 30 %, and up to 50 %) across different PMU measurements and time steps. The IEEE 39-bus and 118-bus datasets each contain 10,000 operating points (OP), incorporating 10 different three-phase fault scenarios with multiple permutations to have a balanced

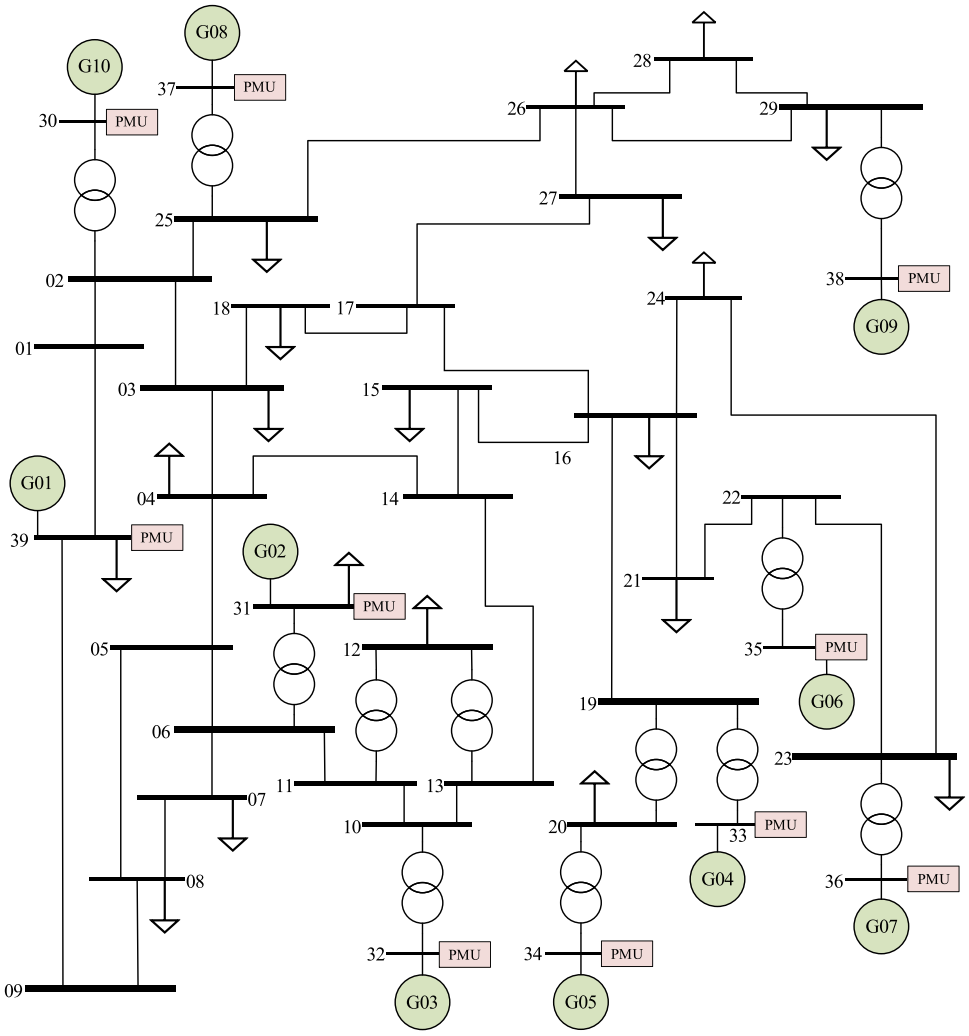


Fig. 4. Single-line diagram of the New England 39-bus test system.

Table 1
Contingencies for the New England 39-bus test system.

Contingency	1	2	3	4	5	6	7	8	9	10
Fault Setting	Fault bus1 Trip 1–39	Fault bus39 Trip 1–39	Fault bus3 Trip 3–4	Fault bus4 Trip 3–4	Fault bus14 Trip 14–15	Fault bus15 Trip 14–15	Fault bus15 Trip 15–16	Fault bus16 Trip 15–16	Fault bus16 Trip 16–17	Fault bus17 Trip 16–17

Table 2
Contingencies for the 118-bus IEEE test system.

Contingency	1	2	3	4	5	6	7	8	9	10
Fault Setting	Fault bus9 Trip 9–10	Fault bus63 Trip 63–64	Fault bus38 Trip 38–65	Fault bus26 Trip 26–30	Fault bus25 Trip 23–25	Fault bus69 Trip 69–70	Fault bus30 Trip 26–30	Fault bus89 Trip 89–92	Fault bus60 Trip 60–61	Fault bus100 Trip 100–103

distribution of secure and insecure OPs.

3.2. Parameter selection

Hyperparameter tuning in deep learning models is often carried out through a **trial-and-error** process, especially in domain-specific applications such as power system security assessment. In this study, we adopted the same empirical tuning approach to determine optimal values for key parameters such as learning rate, batch size, number of epochs, and dropout rate. This method aligns with practices used in several recent works on dynamic security assessment and stability prediction, including [10,20,26], and [4], where trial-and-error was used to balance model accuracy, training stability, and computational efficiency. By iteratively testing different configurations and evaluating validation performance, we ensured that the selected hyperparameters yielded robust and generalizable results for both the GAIN and CNN models.

Hyperparameters are crucial when creating a quality model and

facilitating the learning process. These parameters directly impact the resulting model parameters and should be treated carefully when making any changes. In the process of training the GAIN algorithm, the dataset contains 10,000 operation points, with 80 % (8000) of instances used for training the algorithm and the remaining 20 % (2000) used to test its performance. In this case, the learning rate hyperparameter is set to 0.8, which determines how much the network's weights are adjusted based on the loss gradient. This algorithm uses the Adam optimization algorithm to optimize performance. The training process consists of 3000 epochs with a batch size of 128.

The CNN algorithm in Fig. 2 follows a structured flow encompassing convolution, pooling, dropout, dense, and input/output layers. Activation functions such as ReLU and softmax are employed. The cross-entropy function computes the loss, and Adam's optimization algorithm is utilized for the gradient descent algorithm. The batch size is 64, and training proceeds in 200 epochs. In addition, the weight decay is set at 10^{-6}, and the learning rate and momentum are established at 0.01 and 0.9, respectively.CNN's dataset also contains 10,000 operation points, with 75 % of instances used for the training set and 25 % used for validation.

3.3. GAIN performance evaluation

To evaluate the performance of GAIN algorithms, MAPE was calculated for all PMUs in 5 % and 10 % missing data rates for the 39 bus-ieee systems, as shown in Table 3. The results were compared to [20], and the proposed method outperformed others. The accuracy of the GAN model in filling up missing data is evident from the proximity between the predicted and real values. Furthermore, the computation time per instance is only 1.5 ms, which is insignificant for the online DSA application; quick decision-making is essential in real-time applications like online DSA. The accuracy of the GAIN model in imputing missing data highlights the significance of its low computation time for online DSA applications. It provides insights into its suitability for practical deployment in real-world power systems. To test the effectiveness of the GAIN algorithm on larger systems like the 118-bus IEEE system, we introduced 5 % of missing data to the system and evaluated the MAPE. The results in Table 4 demonstrate that GAIN can effectively recover missing data even in more extensive systems. The average computation time for this process is 2.2 ms. Given its high accuracy and short computation time, this algorithm can be used in online operational calculations.

3.4. The effect of missing data on the performance of the DSA model

The study investigates how the presence of missing data impacts the performance of the DSA model. To establish its superiority, the study rigorously compares the proposed method with three existing state-of-the-art methods, namely, robust RVFL ensemble learning [18] and ELM and RVFL ensemble learning with GAN [20].

Fig. 5 displays the testing results, indicating the percentage of correctly classified testing samples represented by DSA accuracy. Each point on the graph conveys the DSA accuracy, accounting for identical counts of missing PMU data. The results portray the average DSA accuracy for ten three-phase faults, considering 5 % and 10 % missing data rates for each PMU as the quantity of missing data grows and robust ensemble learning experiences a decline in accuracy to varying extents. While ELM and RVFL ensemble learning with GAN performs well in terms of accuracy, the suggested approach can sustain the accuracy of the DSA model even with increasing numbers of missing PMUs. Conversely, the proposed model remains unaffected, mainly with relatively high DSA accuracy. By providing a detailed analysis and comparison with existing methods, the results establish the effectiveness of the proposed model in dynamic security assessment, even in the presence of missing data. An important observation of this method is that none of the previous techniques have specifically examined the impact of missing data from multiple PMUs simultaneously for consecutive seconds. Fig. 1, presented in our study, vividly demonstrates that the CNN algorithm exhibits remarkable accuracy even when confronted with a substantial amount of missing data.

3.5. DSA performance evaluation

To effectively utilize the CNN algorithm, it is crucial to comprehend and assimilate the parameters and weights embedded within the model. These parameters help the model identify patterns and behaviors within the input data, classifying the network's operating points into secure and insecure categories. The convergence speed, accuracy, and loss of training and validation data for the proposed classification models on the New England 39-bus system and IEEE 118-bus system are illustrated in Figs. 6 and 7, respectively. Fig. 6 shows the convergence speed, accuracy, and loss of the training and validation data for the proposed classification models of the New England 39-bus test system. This information is crucial to evaluate the model's ability to learn and generalize from training data to new instances. The figure shows that the DSA model exhibits training and validation accuracy convergence around the 80th epoch. Notably, the training accuracy reaches 100 %, indicating that the model has been effectively trained, and the test accuracy stands at 99.199 %, showcasing the model's ability to generalize well to unseen data.

To evaluate the performance of the perposed DSA method, ADAI is employed. A more excellent value of ADAI indicates a higher accuracy of DSA and a greater robustness against incomplete PMU measurements. Table 6 outlines the accuracy of three distinct methods, demonstrating that the proposed approach exhibits the highest level of robustness in the face of missing data.

The suggested method boasts a substantial computational advantage, efficiently tackling the challenge of filling incomplete PMU data in varying scenarios with just one trained classifier. This contrasts with current methods, which may necessitate up to 2N-1 classifiers for N PMUs. Notably, the proposed approach's efficiency is underscored by the data showcased in Table 5, clearly demonstrating its marked computational superiority over existing methods that require multiple classifiers for DSA training.

To assess how well the proposed method works, we generalized it across different system sizes and complexities; we started our analysis by applying our approach to the 39-bus IEEE system. This helps us to validate and fine-tune these algorithms after optimizing hyperparameters and adjusting the CNN's architectures. Once the results of the 39-bus system were satisfactory, we applied the proposed method to the 118-bus IEEE system. The more extensive system provides a more realistic representation of the power system and introduces additional

Table 3
Filling missing data performance (MAPE under different missing PMU numbers) New England 39-bus test system.

Number of missing PMUs	1	2	3	4	5	6	7	8	9	10	Average Computation Efficiency
Ref [20]	1.33 %	1.42 %	1.56 %	1.73 %	1.79 %	1.84 %	2.01 %	2.06 %	2.25 %	2.28 %	2.24ms
5 % missing rate	0.023 %	0.029 %	0.027 %	0.030 %	0.039 %	0.041 %	0.043 %	0.046 %	0.054 %	0.061 %	1.5ms
10 % missing rate	0.041 %	0.057 %	0.06 %	0.063 %	0.061 %	0.064 %	0.065 %	0.062 %	0.063 %	0.07 %	1.5ms

Table 4
Filling missing data performance (Percentage of MAPE under different missing PMU Numbers) 118-bus IEEE system.

Number of missing PMUs	1	2	3	4	5	6	7	8	9	10
5 % missing rate	0.038	0.049	0.052	0.052	0.053	0.062	0.062	0.065	0.066	0.065
Number of missing PMUs	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**Average Computation Efficiency**
5 % missing rate	0.067	0.070	0.071	0.077	0.080	0.089	0.091	0.091	0.10	2.2ms

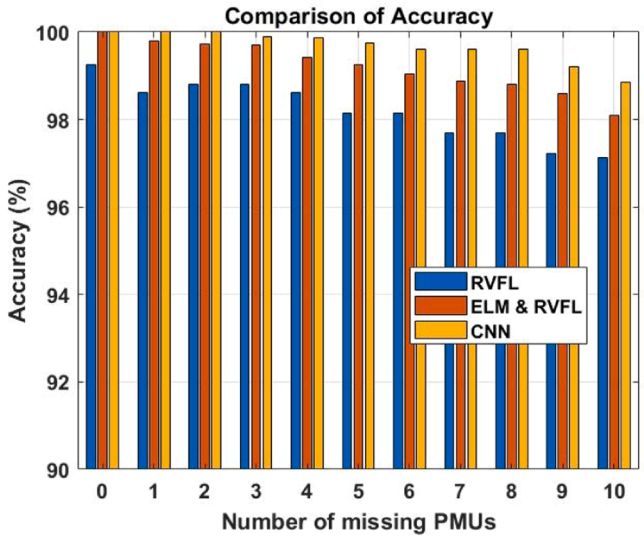


Fig. 5. A comparison of DSA testing results of methods for ten 3-phase faults.

complexity and challenges. The results demonstrate the consistent performance and effectiveness of the 39-bus and IEEE 118-bus systems. It strengthens the credibility and applicability of the approach across various sizes of power systems. The high-precision DSA model proposed in this study has demonstrated its efficacy for application in large power systems. The model exhibits a training accuracy of 98.53 % and a validation accuracy of 96.79 %, shown in Fig. 7, indicating its suitability in complex power system environments. The findings highlight the potential of the proposed method as a reliable and accurate tool for optimal dynamic security assessment of power systems.

3.6. *Sensitivity analysis*

Sensitivity analysis offers decision-makers several advantages. Firstly, it thoroughly examines all variables, leading to more dependable predictions. Secondly, it aids decision-makers in pinpointing future improvements. The sensitivity analysis applied to the rate of missing data for the GAN's model has demonstrated remarkable robustness, even under challenging conditions. The methodology proposed in this study exhibits significantly higher accuracy in filling in missing data than other existing methods. This result underscores the reliability of predictions made using this model. Moreover, sensitivity analysis serves as a valuable tool for identifying potential areas for future enhancement, ultimately boosting the efficacy of the proposed methodology. Table 6 illustrates the changes in MAPE concerning the missing data rate, indicating that the GAN-based data recovery model accurately restores data even with up to a 50 % absence, preserving the DSA model's accuracy at a reasonable level.

4. Conclusion

In conclusion, the proposed approach for DSA using a GAIN-based method and CNN algorithm has shown great performance in

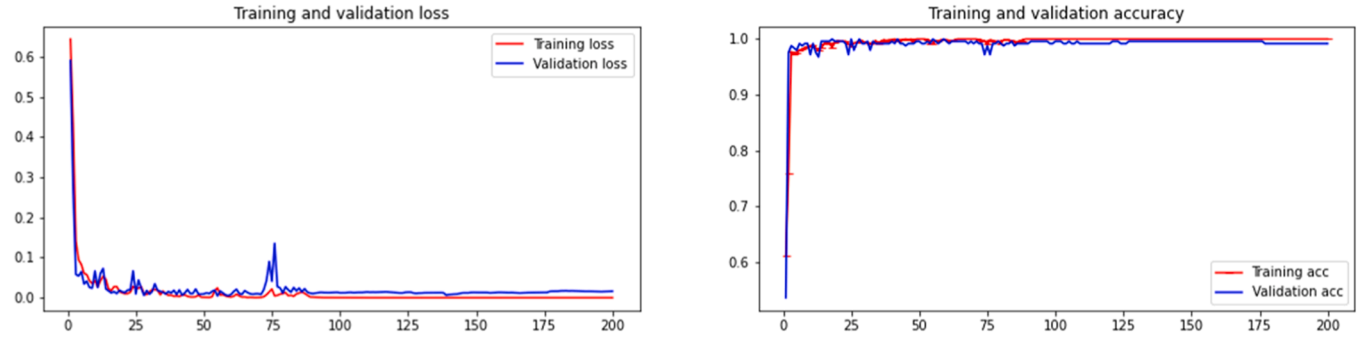


Fig. 6. Training and validation accuracy and loss in 200 epochs for the England 39-bus test system.

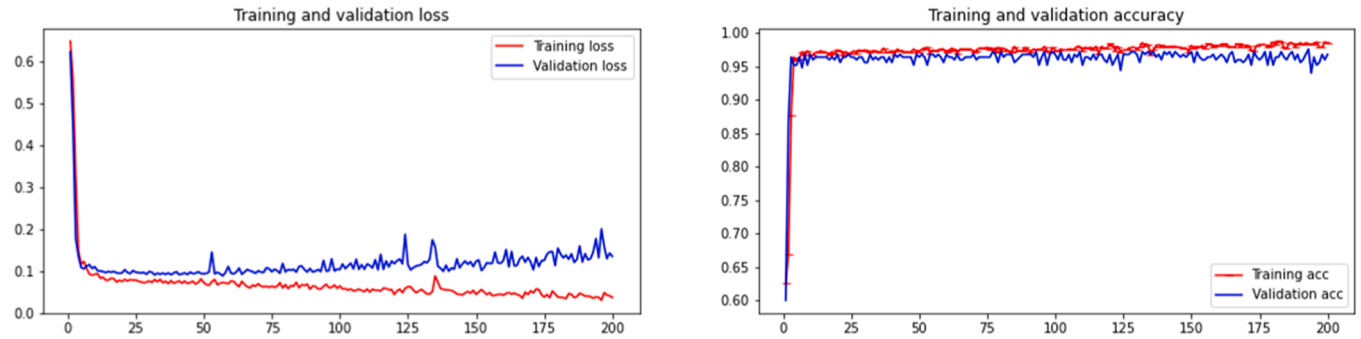


Fig. 7. Training and validation accuracy and loss in 200 epochs for IEEE 118-bus system.

Table 5
Filling missing data performance (MAPE under different missing data rates for all PMUs).

Missing data rate	10 %	20 %	30 %	40 %	50 %	60 %	70 %
MAPE	0.10 %	0.12 %	0.40 %	0.62 %	0.64 %	1.55 %	4.18 %
RMSE	0.0577	0.0542	0.0586	0.0612	0.0839	0.1357	0.2406

Table 6
Comparing the robustness and computational efficiency of the proposed method with previous studies.

method	Computational Efficiency (number of classifiers)	ADAI
Proposed method	1	99.48 %
Ensemble learning ELM & RVFL [20]	1	99.04 %
Robust Ensemble Learning [18]	36	97.96 %

simultaneously addressing the issue of missing data in multiple PMUs. The proposed model has been validated using the IEEE 39-bus network, and the results demonstrate the algorithm's effectiveness in accurately assessing the dynamic security of power systems. Furthermore, the numerical results demonstrate that the GAIN model achieves an imputation accuracy of MAPE $<$ 0.1 % for 10 % missing data, while MAPE is under 0.64 % even up to 50 % data loss. The CNN-based DSA model achieves 99.48 % accuracy on the IEEE 39-bus system and 98.53 % on the IEEE 118-bus system, outperforming traditional ensemble learning approaches while ensuring real-time classification within 3 ms. The approach has consistently performed in the 39-bus and 118-bus power systems, which affirms its robustness and adaptability. This strengthens its credibility and applicability across various power system sizes and underscores its potential for widespread use in dynamic security assessments. The data-driven approach using AI techniques offers a promising and precise means of assessing power system security while mitigating computational burden. The research outcomes are expected to significantly contribute to developing reliable and efficient DSA techniques for power systems.

Future work will focus on enhancing the proposed framework by integrating anomaly detection techniques to improve cyber resilience, enabling the system to adapt to cyber threats and data irregularities. Additionally, we plan to incorporate generator outage scenarios into the DSA framework to improve its ability to generalize across a broader range of operational disturbances. Ultimately, our goal is to deploy the framework in real-world control centers, ensuring seamless integration with existing EMS systems across various network configurations. As part of this effort, we also aim to model cyberattacks and communication disruptions that cause data loss in real-world power systems, including simulations of different attack types and communication failures with varying intensities and frequencies. These enhancements will further strengthen the framework's robustness, adaptability, and reliability in securing modern power grids amid evolving challenges.

CRediT authorship contribution statement

Farzaneh Boronuosi: Writing – original draft, Validation, Software, Formal analysis, Data curation, Conceptualization. **Sasan Azad:** Writing – review & editing, Writing – original draft, Supervision. **Mohammad-Taghi Ameli:** Writing – review & editing, Supervision. **Mohammad Reza Shadi:** Writing – original draft, Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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